Exp: 10
29/8/26

a) Internetworking with Routers

Aim:

To implement internetworking with routers in CISCO Packet Tracer simulator.

Designing and Configuring a simple internetwork:

In this network, a router and 2 PCs are used. Computers are connected with routers using a copper straight-through cable.

Procedure:

Step 1: Configuring Router

i) Select the router and open CLI.

ii) Press enter to start configuring the network

iii) Type enable to activate the privileged mode.

Command for Router

```
# enable
# config t
# interface FastEthernet 0/0
```

ip address 192.168.10.1 255.255.255.0
no shutdown
interface FastEthernet 0/1
ip address 192.168.20.1 255.255.255.0
no shutdown

Step 2 : Configuring PCs :

i) Assign IP address to every PC.

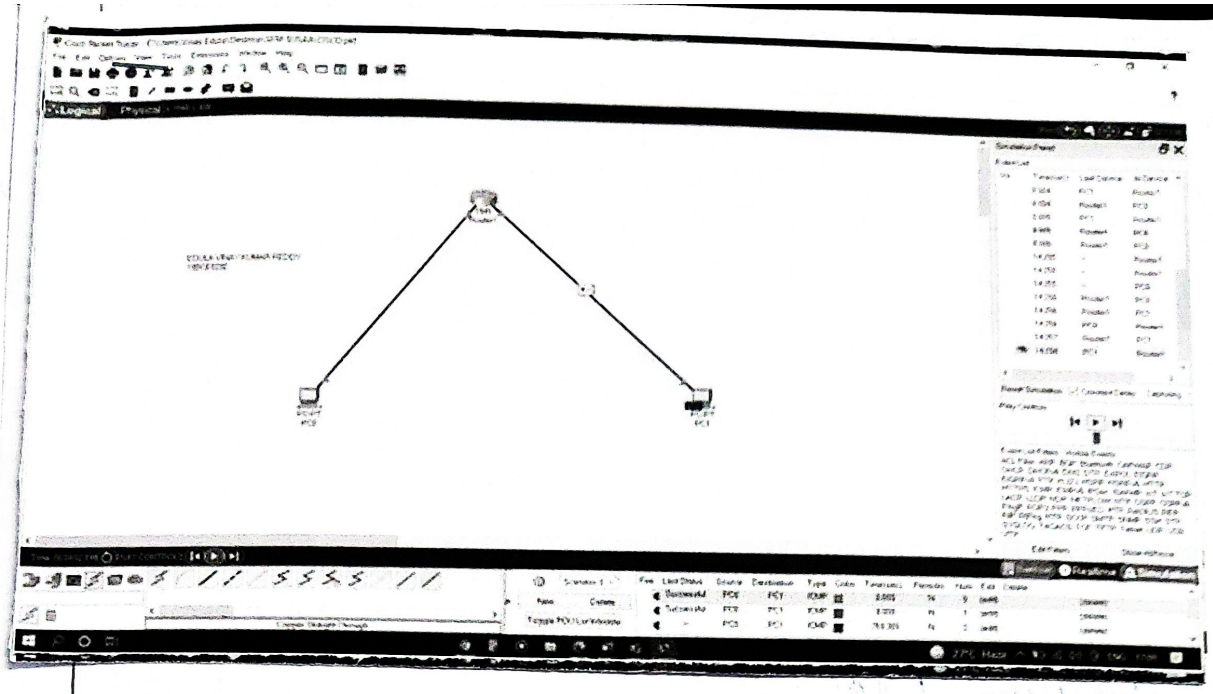
ii) Select the PC, & go to the desktop and select IP Configuration and assign an IP address, Default Gateway, Subnet Mask.

iii) Assign the default gateway of PC0 and PC1 as 192.168.10.1, and 192.168.20.1 respectively.

Step 3 : Connecting PCs with router :

i) Connect the FastEthernet 0 port of PC0 with FastEthernet 0/0 port of Router 1.

ii) Connect the FastEthernet 0 port of PC1 with FastEthernet 0/1 port of Router.



Result:

The implementation of Internetworking with routers has been completed successfully.

Exp:10

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b) Design and Configure an internetwork using wireless router, DHCP server and Internet Cloud.

Aim:

To design and configure an internetwork using wireless router, DHCP server and internet cloud.

Addressing Table

Device	Interface	IP Address	Subnet Mask	Default Gateway
PC	Ethernet 0	DHCP		192.168.0.1
Wireless Router	LAN	192.168.0.1	255.255.255.0	
Wireless Router	Internet	DHCP		
Cisco.com server	Ethernet 0	208.67.220.220	255.255.255.0	
Laptop	Wireless 0	DHCP		

Procedure:

Part 1: Building a simple network

Step 1: Build the topology

* Add network devices using the device selection box into the workspace.

* Change the display name of the devices by clicking

on the device icon and click on the Config tab and type in the new name.

* Add the physical cabling between devices.

Part 2 : Configure the Network Devices

Step 1 : Configure the wireless router

* Click on the wireless router icon and click on GUI tab to view configuration options
* Then click on the Wireless tab, and the only thing that needs to be changed is Network Name (SSID).
* Then save the settings.

Step 2 : Configure the laptop

* Configure the laptop by clicking on its icon and then selecting Physical tab.
* There remove the Ethernet copper module and replace it with the Wireless WPC300N module.

Step 3 : Configure the PC

* Configure the PC for Wired network.

Step 4: Configure the Internet Cloud

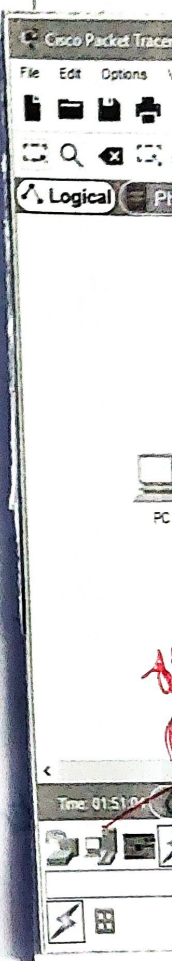
* Click on the internet cloud icon and then click on the Physical tab.
* The cloud device needs 2 modules, PT-CLOUD-NM-1CX and PT-CLOUD-NM-1CFE
* Identify the from and to ports and add them.
* In the Config tab click Ethernet under Interface, select Cable as the Provider.

Step 5: Configure the Cisco.com server

* Configure the Cisco.com server as a DHCP server.
* Configure the server as DNS server to provide domain name to IPv4 address resolution.
* Configure the Global Settings in the Config tab.
* Configure the FastEthernet 0 interface settings in the server.

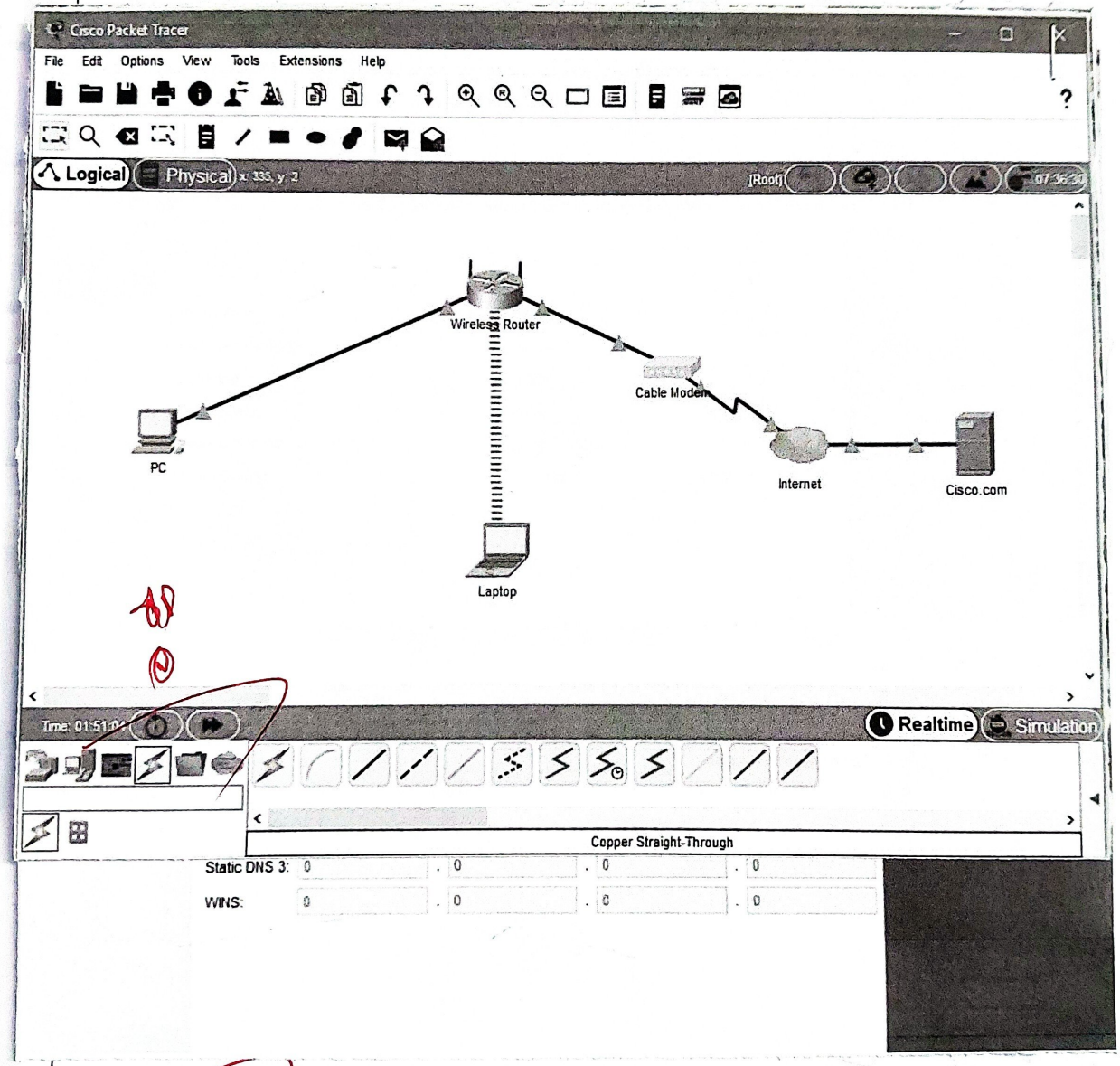
Part 3

* Verify
from
* Test



Part 3: Verify the Connectivity

* Verify that the PC is receiving IPv4 configuration from DHCP.
* Test connectivity to the server from the PC.



Result:

Implementation of Internetworking using DHCP Server, Internet Cloud, and routers has been completed.